

Air Has Weight

Balloon Experiment

Grade 4 | STEM



What Will We Learn?

Today we will discover that air is not nothing – it actually has weight!

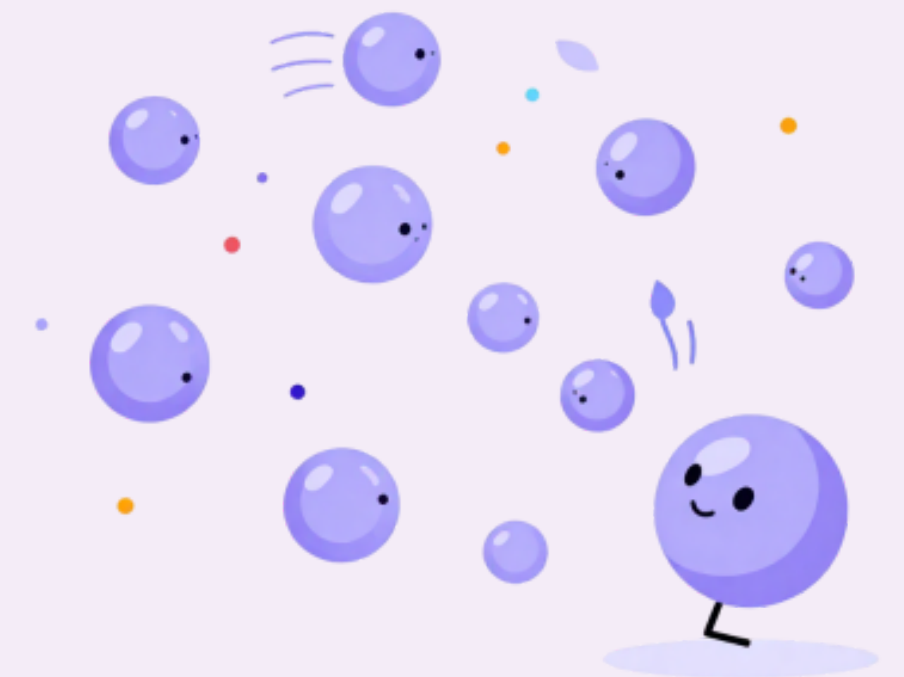
We will prove it with a simple experiment.



What is Air?

Air is a mixture of gases – mostly nitrogen and oxygen.

Even though we cannot see it, air is real matter that takes up space.



Does Air Have Weight?

Think about it – can something you
cannot see still have weight?

Let's do an experiment to find out!



Let's Find Out!

We will use balloons and a simple balance to prove that air has weight.

This is called the Balloon Experiment!



Materials Needed

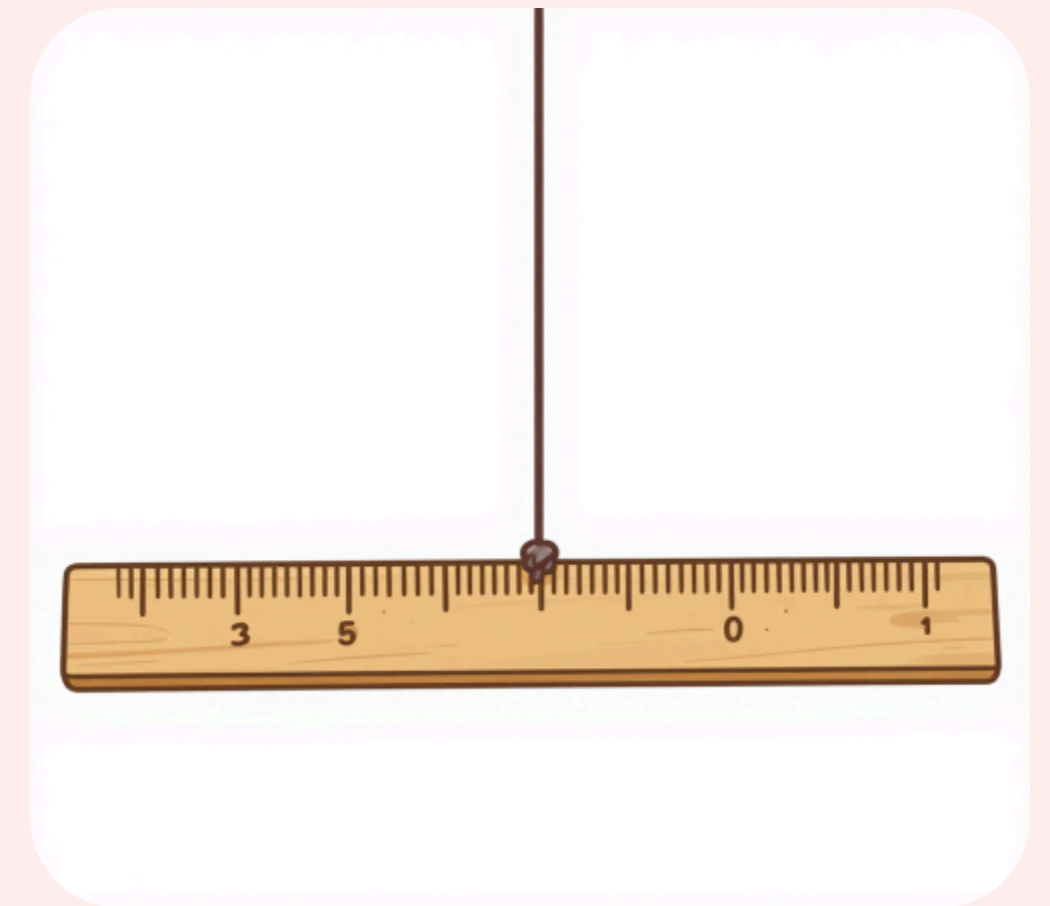
1. A wooden ruler or stick
2. A piece of string
3. Two balloons (same size)
4. Tape
5. A pin or needle



Step 1: Make a Balance

Tie a string at the centre of the ruler.
Hang it from a support (like a chair back).

Make sure it balances perfectly
– stays level and horizontal.



Step 2: Attach Balloons

Blow up both balloons to the same size.
Tape one balloon to each end of the ruler.

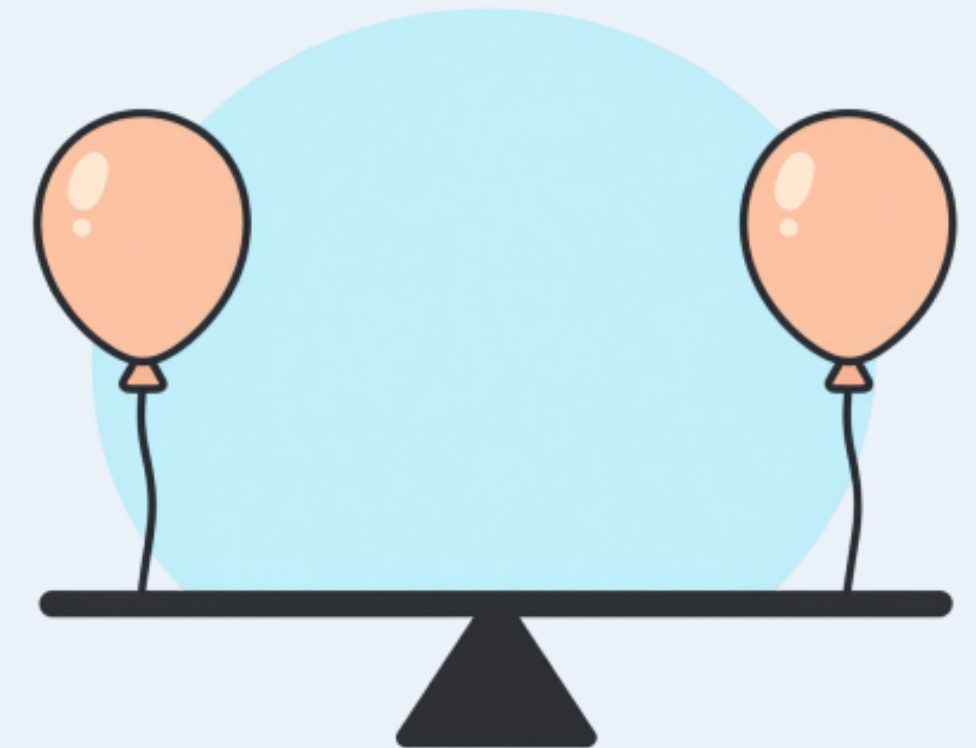
Check again – the ruler should still
be balanced (level).



Step 3: Check the Balance

Both balloons are inflated and equal.
The ruler is perfectly balanced!

This means both sides weigh the
same.



Step 4: Pop One Balloon!

Now carefully pop ONE balloon
using a pin or needle.

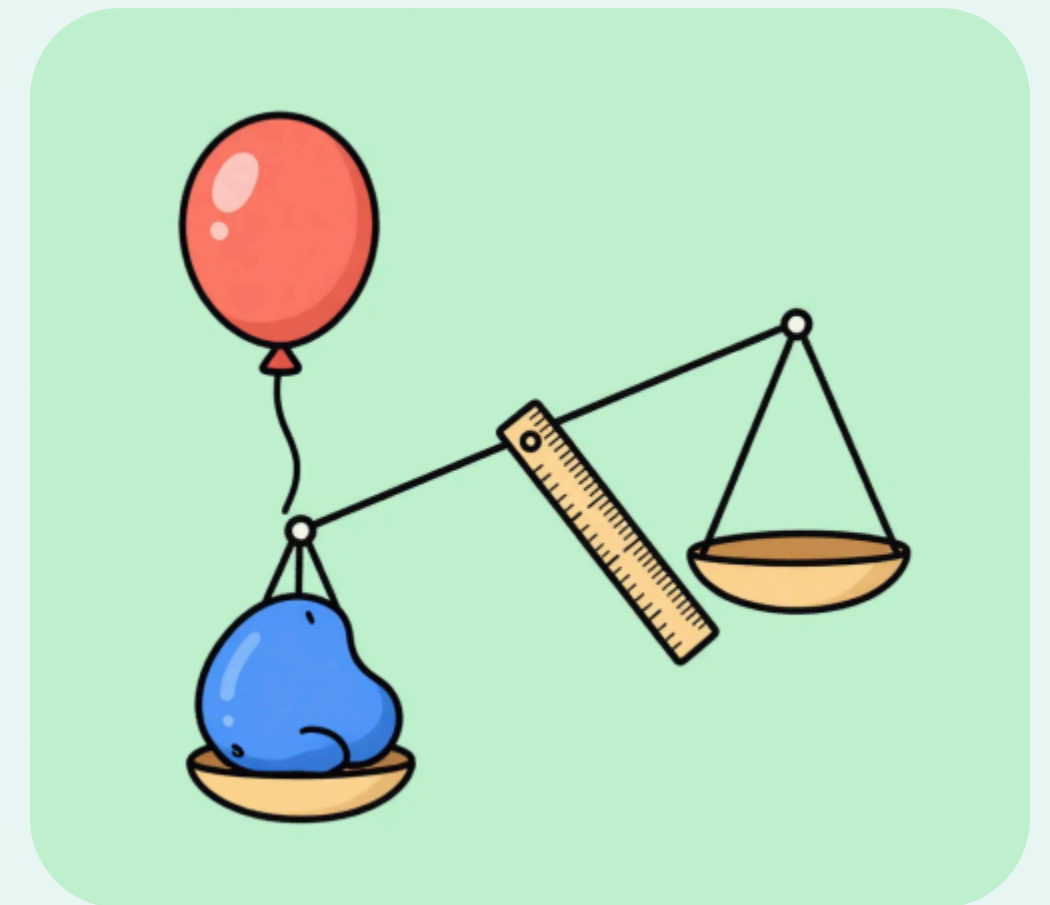
Watch what happens to the
balance!



What Happened?

The side with the inflated balloon
goes DOWN!

The popped balloon side goes UP.
The balance is no longer level.

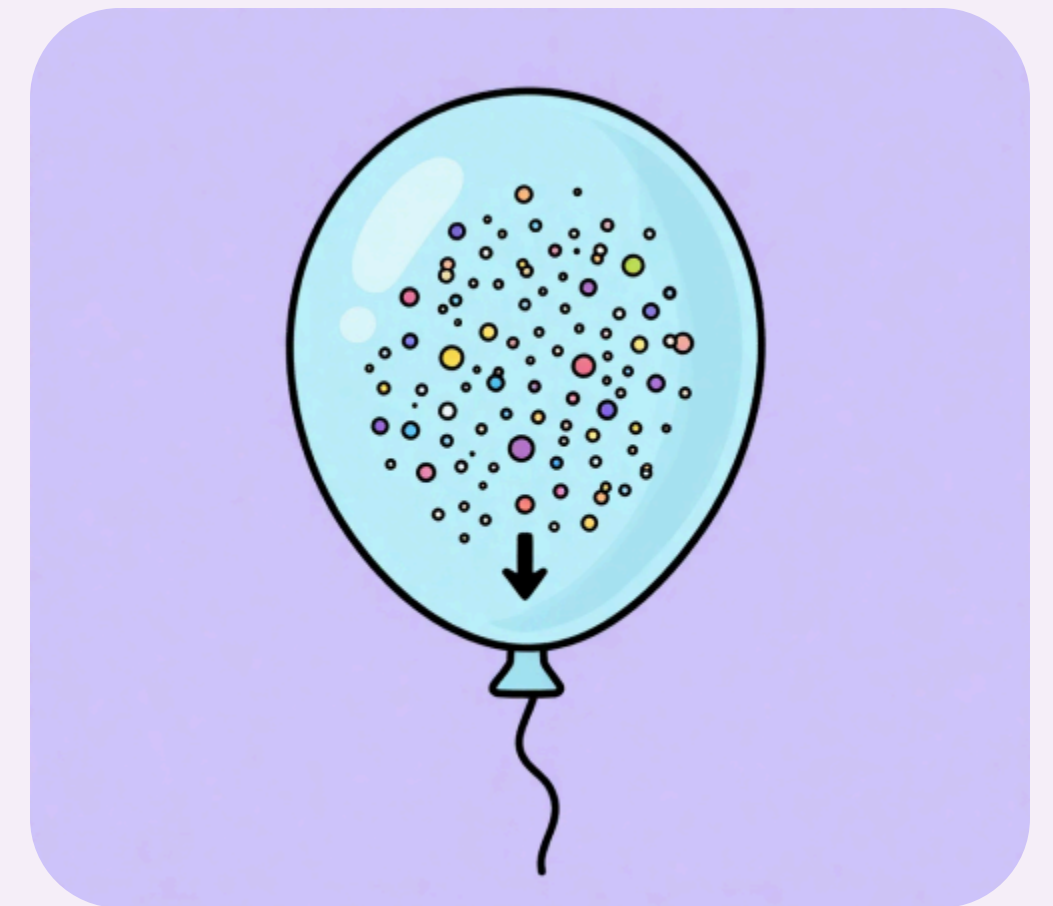


Why Did This Happen?

The inflated balloon has air trapped inside.

That air has weight!

When we pop a balloon, the air escapes and that side becomes lighter.



The Science Behind It

Air is made of tiny gas molecules
(nitrogen, oxygen, and more).

These molecules have mass,
so air has weight!



Real-Life Connection

Tyres are filled with air – that's why a flat tyre feels lighter to lift!

A football with air weighs more than a deflated one.



Key Takeaway

Air has weight!

Even though we can't see it,
air is real matter with mass.
Our experiment proved it!



Great Job, Scientists! 🎉

You proved that air has weight
using a simple balloon experiment!

Keep asking questions and
exploring the world around you!

